

COS1501

STUDY UNIT 6

SPECIAL KINDS OF RELATION

Complete taught notes

Order relations • proof strategies • equivalence relations • equivalence classes • partitions • n-ary relations • functions

What this PDF contains

Full source-aligned coverage of COS1501 Study Unit 6 (pp. 83-102 of the supplied study guide), expanded into lesson-style notes. Every official topic is explained, every major category has worked examples, the main source activities are solved, and the final pages include exam methods, traps, a practice test and a mastery checklist.

Colour key

Green = definitions. Blue = worked examples. Yellow = memory rules. Red = exam traps. Purple = source/teaching notes. Orange = exam method or proof strategy.

Source alignment

Terminology and organization follow the supplied UNISA COS1501 study guide. Added explanations, extra examples and activity solutions are newly written for learning. Where the source appears to contain a typo, it is explicitly flagged rather than silently changed.

Prepared as expanded revision notes for Study Unit 6.

Contents and learning outcomes

By the end of Unit 6 you should be able to do all of the following without guessing:

- Distinguish weak partial, strict partial, weak total/linear and strict total/linear order relations by their required properties.
- Test reflexivity, irreflexivity, symmetry, antisymmetry, transitivity and trichotomy/comparability using formal reasoning.
- Explain why examples can suggest a general result but cannot prove a universal statement, and know when a counterexample or brute-force proof is valid.
- Define and identify equivalence relations, calculate equivalence classes, choose representatives and connect equivalence relations to partitions.
- State and use the four main facts of Theorem 6.1 about equivalence classes.
- Test whether a collection of subsets is a partition and construct the corresponding equivalence relation.
- Recognise binary, ternary and general n-ary relations as subsets of Cartesian products and explain their database use.
- Distinguish a functional relation from a function.
- Determine domain, range and codomain and prove formally whether a relation is a function from A to B.
- Use $f: A \rightarrow B$ and $f(a)=b$ notation correctly and solve the Unit 6 source-style assessment problems.

Part	Coverage
0	Required Unit 5 relation-property recap
1	Order relations
2	Proof strategies
3	Equivalence relations and classes
4	Partitions and relation-partition correspondence
5	n-ary relations
6	Functions
7	Integrated classification method
8	Worked source activities
9	Exam revision, practice test and glossary

The whole unit in one sentence

Unit 6 classifies relations by properties: order relations organise elements, equivalence relations group elements, n-ary relations store multi-field data, and functions enforce exactly one output for every allowed input.

0. Required relation-property recap

Unit 6 builds directly on Unit 5. Before classifying special relations, you must be able to test the ordinary relation properties.

Property	What to check for a relation R on A	Fast failure test
Reflexive	For every $x \in A$, $(x,x) \in R$.	Find one x with $(x,x) \notin R$.
Irreflexive	For every $x \in A$, $(x,x) \notin R$.	Find one x with $(x,x) \in R$.
Symmetric	If $(x,y) \in R$, then $(y,x) \in R$.	Find a pair whose reverse is missing.
Antisymmetric	If $x \neq y$ and $(x,y) \in R$, then $(y,x) \notin R$.	Find distinct x,y with both directions present.
Transitive	If $(x,y), (y,z) \in R$, then $(x,z) \in R$.	Find a two-step chain whose shortcut is missing.
Trichotomy / comparability	For distinct x,y , at least one of xRy or yRx holds.	Find two distinct incomparable elements.

Warm-up example

Let $A=\{1,2,3\}$ and $R=\{(1,1),(2,2),(3,3),(1,2),(1,3),(2,3)\}$. R is reflexive, antisymmetric and transitive. Every two distinct elements are comparable. Therefore this relation will later be classified as a weak total order.

Antisymmetric does NOT mean "not symmetric"

A relation may be both symmetric and antisymmetric. Example: the identity relation $\{(1,1),(2,2)\}$ is symmetric and antisymmetric. Antisymmetry only blocks two-way relations between distinct elements.

1. Order relations (Official 6.1)

An order relation is a relation whose properties let us interpret elements as being arranged “before/after”, “smaller/larger”, “contained in”, “divides”, or in some other consistent ordering. The exact class depends on which properties hold.

1.1 Weak partial order

Definition - weak partial order

A relation R on a set A is a weak partial order iff R is reflexive on A , antisymmetric, and transitive.

Memory rule

WEAK partial = $R \wedge T$: Reflexive + Antisymmetric + Transitive.

Worked example - subset relation on a small set

Let $A = \{\{a\}, \{a, b\}\}$. Define $B \leq C$ iff $B \subseteq C$. Then $S = \{\{\{a\}, \{a\}\}, \{\{a\}, \{a, b\}\}, \{\{a, b\}, \{a, b\}\}\}$. Reflexive: each set is a subset of itself. Antisymmetric: $\{a\} \subseteq \{a, b\}$ but $\{a, b\} \not\subseteq \{a\}$. Transitive: whenever $B \subseteq C$ and $C \subseteq D$, $B \subseteq D$. Hence S is a weak partial order.

Worked example - “is a multiple of” on $\mathbb{Z}^+$

Define xRy iff $x = k \cdot y$ for some $k \in \mathbb{Z}^+$. Reflexive: $x = 1 \cdot x$. Antisymmetric: if $x = my$ and $y = kx$ with positive integers m, k , then $x = (mk)x$, forcing $mk = 1$ and therefore $x = y$. Transitive: $x = my$ and $y = kz$ imply $x = (mk)z$. Thus R is a weak partial order.

Added example - divisibility in the opposite direction

On $\mathbb{Z}^+$, define xDy iff x divides y . Reflexive: $x | x$. Antisymmetric: $x | y$ and $y | x$ force $x = y$ for positive integers. Transitive: $x | y$ and $y | z$ imply $x | z$. So D is a weak partial order, but not a total order because 2 and 3 are incomparable.

Activity 6-4 - weak partial orders (worked)

(a) Membership relation

$A = \{a, b, \{a, b\}\}$, and $c \leq b$ iff $c \in b$. The relation contains $(a, \{a, b\})$ and $(b, \{a, b\})$, but it is not reflexive: for example $(a, a) \notin S$. Therefore it is not a weak partial order.

(b) $x+y$ is even

On $\mathbb{Z}$, xRy iff $x+y$ is even. It is reflexive because $2x$ is even, but it is not antisymmetric: $1R3$ and $3R1$ while $1 \neq 3$. Therefore it is not a weak partial order.

(c) Lexicographic-style order on $\mathbb{Z} \times \mathbb{Z}$

Define $(a, b)R(c, d)$ iff $a < c$, or $a = c$ and $b \leq d$. It is reflexive, antisymmetric and transitive. In fact every two pairs are comparable, so it is a weak total order and therefore also a weak partial order.

1.2 Strict partial order

Definition - strict partial order

For COS1501, a relation R on A is a strict partial order iff R is irreflexive, antisymmetric, and transitive.

Memory rule

STRICT partial = I A T: Irreflexive + Antisymmetric + Transitive.

Use the COS1501 definition in this module

Other textbooks sometimes define a strict partial order using only irreflexivity and transitivity. For this unit, use the study-guide test explicitly: irreflexive + antisymmetric + transitive.

Worked example - "less than" on $\{1,2,3\}$

$S = \{(1,2), (1,3), (2,3)\}$. No loop occurs, so S is irreflexive. No reverse pair occurs for distinct elements, so S is antisymmetric. The only nontrivial chain $1 \rightarrow 2 \rightarrow 3$ has the required shortcut $1 \rightarrow 3$, so S is transitive. Hence S is a strict partial order. Because every two distinct elements are comparable, it is actually a strict total order.

Added example - proper subset

On $P(\{1,2\})$, define $A < B$ iff $A \subset B$ (proper subset). No set is a proper subset of itself, so the relation is irreflexive. It is antisymmetric and transitive. It is strict partial, but not total because $\{1\}$ and $\{2\}$ are incomparable.

Activity 6-5 - strict partial orders (worked)

(a) $S = \{(a, \{a\}), (a, \{b\})\}$

On $A = \{a, \{a\}, \{b\}\}$, there are no loops, no reversed distinct pair, and no two-step chain that creates a missing shortcut. Therefore S is irreflexive, antisymmetric and transitive: a strict partial order. It is not total because $\{a\}$ and $\{b\}$ are incomparable.

(b) Compare only first coordinates

On $Z \times Z$ define $(a,b)R(c,d)$ iff $a < c$. The relation is irreflexive, antisymmetric and transitive, so it is a strict partial order. It is not total: $(1,2)$ and $(1,3)$ are distinct but neither is related to the other because their first coordinates are equal.

1.3 Why "partial"? Comparability and trichotomy

"Partial" means the order may leave some pairs of distinct elements incomparable. The subset relation is the standard example.

Power-set example from the guide

Let $U = \{1,2\}$. $P(U) = \{\subseteq, \{1\}, \{2\}, \{1,2\}\}$. Under $\subseteq$, we have $\subseteq \{1\}$, $\subseteq \{2\}$, $\{1\} \subseteq \{1,2\}$, and $\{2\} \subseteq \{1,2\}$. But $\{1\} \not\subseteq \{2\}$ and $\{2\} \not\subseteq \{1\}$. Thus $\{1\}$ and $\{2\}$ are incomparable.

Comparable means one direction works

For distinct x and y : comparable means xRy or yRx . Trichotomy in this unit is the condition that every pair of distinct elements is comparable.

Visual picture - partial versus total

Relation	Pair chosen	Comparison result
$\subseteq$ on $P(\{1,2\})$	$\{1\}$ and $\{2\}$	Neither $\{1\} \subseteq \{2\}$ nor $\{2\} \subseteq \{1\} \rightarrow$ incomparable
$\leq$ on $\mathbb{Z}$	-7 and 4	$-7 \leq 4 \rightarrow$ comparable
$<$ on $\mathbb{Z}$	9 and 3	$3 < 9 \rightarrow$ comparable in the reverse direction

1.4 Total (linear) orders

Definition - total / linear order

A total (linear) order is a partial order with the additional comparability/trichotomy requirement: every two distinct elements can be compared in at least one direction.

Weak total order - $\leq$ on $\mathbb{Z}$

$\leq$ is reflexive, antisymmetric and transitive. For any integers m, n , either $m \leq n$ or $n \leq m$. Therefore $\leq$ is a weak total order.

Strict total order - $<$ on $\mathbb{Z}$

$<$ is irreflexive, antisymmetric and transitive, and every two distinct integers are comparable. Therefore $<$ is a strict total order.

Added example - alphabetical order

If a finite list of distinct words is ordered lexicographically (dictionary order), every two words can be compared. This is the computer-science reason total orders are useful for sorting.

1.5 Master table of the four order types

Kind of order	Reflexive	Irreflexive	Antisymmetric	Transitive	Trichotomy
Weak partial	Yes	No	Yes	Yes	Not required
Strict partial	No	Yes	Yes	Yes	Not required
Weak total / linear	Yes	No	Yes	Yes	Yes
Strict total / linear	No	Yes	Yes	Yes	Yes

One switch separates weak from strict

Weak orders contain the self-pairs (reflexive). Strict orders exclude all self-pairs (irreflexive). Adding trichotomy upgrades partial $\rightarrow$ total.

1.6 Activity 6-7 - full worked classification

Question 1 - all strict partial orders on $X=\{a,b,c\}$

There are 19 strict partial orders. They can be organised by number of ordered pairs. This is a brute-force proof because X is finite and small.

0 pairs: $\emptyset$

1 pair (6): $\{(a,b)\}, \{(a,c)\}, \{(b,a)\}, \{(b,c)\}, \{(c,a)\}, \{(c,b)\}$

2 pairs (6): $\{(a,b),(a,c)\}, \{(b,a),(b,c)\}, \{(c,a),(c,b)\},$
 $\{(a,b),(c,b)\}, \{(a,c),(b,c)\}, \{(b,a),(c,a)\}$

3 pairs (6):

$\{(a,b),(b,c),(a,c)\}$

$\{(a,c),(c,b),(a,b)\}$

$\{(b,a),(a,c),(b,c)\}$

$\{(b,c),(c,a),(b,a)\}$

$\{(c,a),(a,b),(c,b)\}$

$\{(c,b),(b,a),(c,a)\}$

Which of the 19 are linear?

Exactly the six 3-pair relations are strict total/linear orders. Each corresponds to one possible ordering (permutation) of a,b,c . The empty, 1-pair and 2-pair orders leave at least one pair incomparable.

Question 2 - classify each relation

Case	Classification	Reason
(a) $X=\{\emptyset,\{0\},\{2\}\}$, $R=\{(\emptyset,\{0\}),(\emptyset,\{2\})\}$	Strict partial	Irreflexive, antisymmetric, transitive; $\{0\}$ and $\{2\}$ incomparable.
(b) $X=\{\emptyset,\{\emptyset\},\{\{\emptyset\}\}\}$, $R=\subseteq$	Weak partial	Subset is reflexive, antisymmetric, transitive; $\{\emptyset\}$ and $\{\{\emptyset\}\}$ are incomparable.
(c) $X=\mathbb{Z}$, $R=\leq$	Weak total	Reflexive, antisymmetric, transitive, all integers comparable.
(d) $X=\mathbb{Z}$, $R=>$	Strict total	Irreflexive, antisymmetric, transitive, all distinct integers comparable.
(e) $X=\mathbb{Z}^+$, xRy iff x divides y	Weak partial	Reflexive, antisymmetric, transitive; e.g. 2 and 3 incomparable.

2. Proof strategies (Official 6.2)

Unit 6 repeatedly asks whether a relation has a property for all elements. Your proof method matters as much as the final yes/no answer.

2.1 Examples are useful - but they do not prove “for all”

Main proof rule

A handful of examples cannot prove a universal statement such as “for all $x...$ ” or “for all $x,y...$ ”. They can help you see the pattern, but a general proof must use arbitrary elements.

Getting a “gut feel”

For T on Z defined by xTy iff $x-y=3k$ for some $k \in Z$, examples such as $(10,7)$, $(80,50)$, $(28,73)$, $(-9,-3)$, and $(2,2)$ all belong because each difference is a multiple of 3. These examples suggest what the relation means: x and y have the same remainder modulo 3.

2.2 General / abstract proof

Template

1. Take arbitrary elements satisfying the hypothesis. 2. Translate relation membership into the defining algebraic condition. 3. Manipulate that condition generally. 4. Translate the result back into relation language.

Prove T is reflexive

For arbitrary $x \in Z$, $x-x=0=3 \cdot 0$, where $0 \in Z$. Hence $(x,x) \in T$ for every $x \in Z$. Therefore T is reflexive.

Added example - transitivity of congruence mod 3

Suppose xTy and yTz . Then $x-y=3m$ and $y-z=3n$ for integers m,n . Add: $x-z=(x-y)+(y-z)=3(m+n)$. Since $m+n \in Z$, xTz . Thus T is transitive.

2.3 Counterexample

Counterexample

One valid case that violates a universal claim is enough to prove that the claim is false.

Guide example - trichotomy fails for T

Take 1 and 5. $1-5=-4$ and $5-1=4$; neither is divisible by 3. So neither $1T5$ nor $5T1$. This single incomparable pair proves T does not satisfy trichotomy.

Proof asymmetry

To prove “for all”, you need a general argument. To disprove “for all”, one counterexample is enough.

2.4 Brute force

Brute-force proof

When the set is small and finite, list all relevant possibilities and check every one. Activity 6-7(1), where all strict partial orders on $\{a,b,c\}$ are listed, is the unit's main example.

When brute force becomes a bad idea

Do not try to brute-force an infinite set such as $\mathbb{Z}$. Switch to arbitrary variables and algebraic reasoning.

2.5 Reductio ad absurdum / proof by contradiction

Method

Assume temporarily the opposite of what you want to prove. Derive an impossibility or contradiction. Since the temporary assumption is the weak link, reject it and conclude the desired statement.

Unit 6 use of contradiction

In Theorem 6.1(iv), start with distinct equivalence classes $[x] \neq [y]$. To prove they are disjoint, temporarily assume $[x] \cap [y] \neq \emptyset$. Then some v lies in both classes, giving xRv and yRv . Symmetry gives vRy ; transitivity gives xRy ; Theorem 6.1(ii) then gives $[x] = [y]$, contradicting $[x] \neq [y]$. Hence the intersection must be empty.

Contradiction skeleton

Goal P . Assume $\neg P$. Derive C and $\neg C$ (or another impossibility). Therefore $\neg P$ is false, so P is true.

3. Equivalence relations (Official 6.3)

Definition - equivalence relation

A relation R on a set A is an equivalence relation iff R is reflexive on A , symmetric, and transitive.

Memory rule

Equivalence = R S T: Reflexive + Symmetric + Transitive.

Equivalence relations are used when we want to ignore irrelevant differences and group objects that count as “the same” for one chosen characteristic.

Guide idea - households

If xRy means “ x slept under the same roof as y on census night”, then (under the guide’s simplifying assumption) the relation groups people into households. Each household becomes one equivalence class.

Guide idea - grade bands

Students with different raw marks can still be equivalent under the relation “received the same grade band”. All E-students form one class, all D-students another, and so on.

Added example - congruence modulo 4

On $\mathbb{Z}$, define xRy iff $x - y$ is divisible by 4. Integers are grouped according to remainder 0, 1, 2 or 3 when divided by 4. This relation is reflexive, symmetric and transitive, hence an equivalence relation.

3.1 Equivalence classes

Definition - equivalence class

For $x \in A$, the equivalence class of x is $[x] = \{y \mid y \in A \text{ and } xRy\}$. It contains exactly the elements equivalent to x .

Parity relation on $\mathbb{Z}$

Let xRy iff $y - x$ is even. Then $[0] = \{\dots, -4, -2, 0, 2, 4, \dots\}$, the even integers, and $[1] = \{\dots, -3, -1, 1, 3, 5, \dots\}$, the odd integers. There are only two distinct classes.

Same class, many names

$[0] = [2] = [-4] = \dots$ and $[1] = [3] = [-1] = \dots$. A different representative can name the same equivalence class.

3.2 Rational numbers as equivalence classes of quotients

The guide uses $a/b R c/d$ iff $ad - bc = 0$. This says the cross-products are equal, so the two quotients represent the same rational number.

Example

$1/3 R 2/6$ because $1 \cdot 6 - 3 \cdot 2 = 0$. Likewise $2/5 R 20/50$ because $2 \cdot 50 - 5 \cdot 20 = 0$.

Activity 6-8 - prove the quotient relation is an equivalence relation

Reflexive

For any a/b , $ab - ba = 0$. Therefore $a/b \ R \ a/b$.

Symmetric

If $a/b \ R \ c/d$, then $ad - bc = 0$. Hence $cb - da = -(ad - bc) = 0$, so $c/d \ R \ a/b$.

Transitive

Suppose $a/b \ R \ c/d$ and $c/d \ R \ e/f$. Then $ad = bc$ and $cf = de$. Multiply the first equality by f : $adf = bcf$. Replace cf by de : $adf = bde$. Since $d \neq 0$, cancel d to get $af = be$. Therefore $a/b \ R \ e/f$. So R is transitive.

Conclusion

R is reflexive, symmetric and transitive, so it is an equivalence relation. The class $[1/3]$ contains $1/3, 2/6, 3/9, 4/12, \dots$.

3.3 Representatives and modulo classes

A representative is one chosen element used to name a class. The guide convention is to use a convenient smallest non-negative representative when possible.

Modulo 3

Under congruence modulo 3, the classes are $[0]$, $[1]$ and $[2]$. Every integer belongs to exactly one of them according to its remainder when divided by 3.

Class	Some members	Meaning
$[0]$	$\dots, -6, -3, 0, 3, 6, \dots$	Multiples of 3
$[1]$	$\dots, -5, -2, 1, 4, 7, \dots$	Remainder 1 mod 3
$[2]$	$\dots, -4, -1, 2, 5, 8, \dots$	Remainder 2 mod 3

3.4 Activity 6-10 - equivalence relations (worked)

Question 1 - all equivalence relations on $\{a, b, c\}$

Equivalence relations correspond exactly to partitions. A 3-element set has five partitions, so it has five equivalence relations.

- $\{(a, a), (b, b), (c, c)\}$
- $X \times X$ (all 9 ordered pairs)
- $\{(a, a), (b, b), (c, c), (a, b), (b, a)\}$ classes $\{a, b\} \mid \{c\}$
- $\{(a, a), (b, b), (c, c), (a, c), (c, a)\}$ classes $\{a, c\} \mid \{b\}$
- $\{(a, a), (b, b), (c, c), (b, c), (c, b)\}$ classes $\{b, c\} \mid \{a\}$

Question 2

Case	Equivalence?	Classes
(a) $R = \{(a,a), (b,b), (c,c)\}$	Yes	$\{a\}, \{b\}, \{c\}$
(b) $R = X \times X$	Yes	One class: $\{a,b,c\}$
(c) $C \subset R \subset D$ iff $C \cap \{2\} = D \cap \{2\}$ on $P(\{1,2,3\})$	Yes	Class without 2: $\emptyset, \{1\}, \{3\}, \{1,3\}$. Class with 2: $\{2\}, \{1,2\}, \{2,3\}, \{1,2,3\}$.

Question 3 - $x-y$ is a multiple of 4

Property	Result	Reason / counterexample
Reflexive	Yes	$x-x=0$ is a multiple of 4.
Irreflexive	No	Reflexivity already gives every loop.
Symmetric	Yes	If $x-y=4k$, then $y-x=4(-k)$.
Antisymmetric	No	$1R5$ and $5R1$, but $1 \neq 5$.
Transitive	Yes	$(x-y)+(y-z)=x-z$, sum of multiples of 4.
Trichotomy	No	1 and 2 are incomparable modulo 4.

Classification and classes

R is an equivalence relation. Its four classes are $[0], [1], [2], [3]$, the residue classes modulo 4.

Question 4 - positive rational scaling

On $\mathbb{Q}^+$, xRy iff $y=(a \cdot x)/b$ for some $a, b \in \mathbb{Z}^+$.

Proof

Reflexive: choose $a=b=1$. Symmetric: if $y=ax/b$, then $x=by/a$. Transitive: if $y=ax/b$ and $z=cy/d$, then $z=(ca)x/(db)$. Hence R is reflexive, symmetric and transitive.

Equivalence classes

For any positive rationals x and y , y/x is itself a positive rational a/b , so xRy . Thus the relation is actually universal on $\mathbb{Q}^+$: every element is equivalent to every other, and there is one class $\mathbb{Q}^+$.

Question 5 - symmetry iff $R=R^{-1}$

Proof

($\Rightarrow$) If R is symmetric and $(x,y) \in R$, then $(y,x) \in R$, which means $(x,y) \in R^{-1}$. Thus $R \subseteq R^{-1}$. The same argument gives $R^{-1} \subseteq R$, so $R=R^{-1}$. ($\Leftarrow$) If $R=R^{-1}$ and $(x,y) \in R$, then $(x,y) \in R^{-1}$, so $(y,x) \in R$. Therefore R is symmetric.

3.5 Theorem 6.1 - what equivalence classes must do

Theorem 6.1

If R is an equivalence relation on A , then: (i) $x \in [x]$ for every $x \in A$; (ii) $xRy \Rightarrow [x] = [y]$; (iii) $[x] = [y] \Rightarrow xRy$; (iv) either $[x] = [y]$ or $[x] \cap [y] = \emptyset$.

(i) Every element belongs to its own class

Proof

R is reflexive, so xRx . By the definition of $[x]$, this means $x \in [x]$.

(ii) Equivalent elements have the same class

Proof

Assume xRy . Let $w \in [x]$, so xRw . By symmetry, wRx . With xRy , transitivity gives wRy . By symmetry, yRw , so $w \in [y]$. Thus $[x] \subseteq [y]$. Reversing the roles of x and y gives $[y] \subseteq [x]$. Hence $[x] = [y]$.

(iii) Same class implies equivalent representatives

Proof

Assume $[x] = [y]$. By (i), $y \in [y]$. Since $[x] = [y]$, $y \in [x]$. By definition of $[x]$, xRy .

(iv) Distinct equivalence classes are disjoint

Proof by contradiction

Assume $[x] \neq [y]$ but $[x] \cap [y] \neq \emptyset$. Choose v in the intersection. Then xRv and yRv . Symmetry gives vRy , so transitivity gives xRy . By (ii), $[x] = [y]$, contradiction. Therefore if the classes are not equal, their intersection is empty.

The key consequence

Equivalence classes never partially overlap. Two classes are either exactly the same set or completely disjoint.

4. Partitions

Equivalence relations and partitions are two views of the same grouping structure: a relation tells you which elements are equivalent; a partition shows the groups directly.

Definition - partition

For a nonempty set A , a partition is a set $S=\{S_1, S_2, S_3, \dots\}$ of subsets of A such that: (1) every part S_i is nonempty; (2) different parts are disjoint; (3) the union of all parts is A .

Partition test

NO EMPTY PARTS + NO OVERLAP + NOTHING MISSING.

Valid partition

$A=\{a,2,b\}$. The intended partition in the guide's explanation is $\{\{a,2\},\{b\}\}$: both parts are nonempty, their intersection is $\emptyset$, and their union is A .

Source typo flagged

The source line displays $\{\{a,2\},\{3\}\}$ even though $A=\{a,2,b\}$; the immediately following explanation uses $\{b\}$. This PDF follows the explanation and treats $\{b\}$ as the intended second part.

Invalid partition - missing element

$\{\{a\},\{b\}\}$ is not a partition of $\{a,2,b\}$ because 2 belongs to no part.

Activity 6-11 - invalid because of overlap

$\{\{goat,pig\},\{pig,cow\}\}$ is not a partition of $\{goat,pig,cow\}$: the two parts intersect in $\{pig\}$, so an element would belong to two parts.

4.1 From a partition to an equivalence relation

Partition $\{\{1,2\},\{3\}\}$

Elements in the same part are equivalent. Therefore $R=\{(1,1),(1,2),(2,1),(2,2),(3,3)\}$. The class $[1]=[2]=\{1,2\}$, and $[3]=\{3\}$.

4.2 Theorem 6.2

Theorem 6.2

Suppose A is nonempty and partitioned. Define xRy iff x and y belong to the same partitioning subset. Then R is an equivalence relation on A .

Proof

Reflexive: every x lies in the same part as itself. Symmetric: if x and y are in the same part, then y and x are in that same part. Transitive: if x and y share a part and y and z share a part, the partition condition says y cannot belong to two different parts; therefore x,y,z are in the same part, so xRz . Hence R is reflexive, symmetric and transitive.

Two-way correspondence

Equivalence relation $\rightarrow$ its equivalence classes form a partition. Partition $\rightarrow$ "same part" defines an equivalence relation.

4.3 Activity 6-12 - partitions (worked)

Case	Partition?	Reason / corresponding relation
(a) $\{\emptyset, \{1\}, \{2, 3\}\}$	No	A partition may not contain an empty part.
(b) $\{\{1\}, \{2\}, \{1, 3\}\}$	No	$\{1\}$ overlaps $\{1, 3\}$.
(c) $\{\{1, 3\}, \{2\}\}$	Yes	R has pairs $(1, 1), (1, 3), (3, 1), (3, 3), (2, 2)$.
(d) $\{\{1\}, \{2\}\}$	No	3 is missing from the union.
(e) $\{\{0\}, \mathbb{Z}^+, \text{Neg}\}$ on $\mathbb{Z}$	Yes	xRy iff x, y are both 0, both positive, or both negative.
(f) $\{\{0\}, \{1\}, \{2\}, \{3\}, \{4\}\} \bmod 5$	Yes	xRy iff $x - y$ is divisible by 5; the parts are the five residue classes.

4.4 Why equivalence relations and partitions matter in computing

- The guide points forward to Boolean algebra, where equivalence-like grouping ideas recur.
- Finite-state-machine minimisation groups states that behave equivalently, reducing complexity and cost.
- In software and data modelling, equivalence classes let us treat many distinct representations as the same abstract value.

Computing intuition

Two machine states can have different names but be equivalent if no future input can distinguish their behaviour. Minimisation then keeps one representative state for the whole class.

5. n-ary relations (Official 6.4)

5.1 From ordered pairs to ordered n-tuples

Ordered n-tuple

For $n \geq 2$, $(x_1, x_2, \dots, x_n)$ is an ordered n-tuple. Coordinate position matters: x_1 is first, x_2 second, ..., x_n nth.

Examples

$(1, 0, 0)$ is a 3-tuple. (student, module, mark, year) is a 4-tuple. Reordering coordinates changes the meaning.

5.2 Binary, ternary and general n-ary relations

n-ary relation

A binary relation consists of ordered pairs. A 3-ary/ternary relation consists of triples. A 4-ary/quaternary relation consists of 4-tuples. In general, an n-ary relation R is a set of n-tuples and can be viewed as $R \subseteq A_1 \times A_2 \times \dots \times A_n$.

"Arity" = number of coordinates

2-ary = binary, 3-ary = ternary, 4-ary = quaternary, n-ary = n coordinates.

Small database example

Let $\text{Student} = \{\text{Amina}, \text{Bilal}\}$, $\text{Module} = \{\text{COS1501}, \text{COS1511}\}$, $\text{Mark} = \{0, \dots, 100\}$. A 3-ary relation might contain (Amina, COS1501, 78) and (Bilal, COS1511, 65). Each tuple stores one complete record across three fields.

5.3 Guide database example

The study guide models a university record using ten sets: valid student names, student numbers, dates of birth, sex values, addresses, modules, credits, balance brought forward, paid-to-date values and amount owing.

10-ary record

A tuple such as (name, student number, date of birth, sex, address, module, credits, balance, paid, amount owing) is one element of $X_1 \times X_2 \times \dots \times X_{10}$. The set of all valid records forms a 10-ary relation.

Projection to fewer fields

If an application only needs name, sex and credits, use triples from $X_1 \times X_4 \times X_7$. This produces a 3-ary relation instead of carrying all ten fields.

Relational database connection

This is the conceptual basis of the relational database model: data are organised as tuples so records can be stored, queried and modified systematically. Later database modules develop this further.

6. Functions (Official 6.5)

A function is a special binary relation. The key is not merely “one output”; the domain must also be exactly the declared input set.

6.1 Functional relation

Definition - functional relation

Suppose $R \subseteq X \times Y$. R is functional iff whenever $(x,y) \in R$ and $(x,z) \in R$, then $y=z$. In plain language: a first coordinate that appears in the relation is paired with only one second coordinate.

Functional = no input splits

An input is allowed to be missing entirely. But if it appears, it cannot point to two different outputs.

Functional but incomplete

From $\{1,2,3\}$ to $\{a,b,c\}$, $S = \{(1,a), (2,c)\}$ is functional. Inputs 1 and 2 each have one output. Input 3 is absent, so S may fail to be a function even though it is functional.

Formula relation

On Z , define $y=x+5$. If (x,y) and (x,z) both satisfy the rule, then $y=x+5=z$. Therefore the relation is functional.

Not functional

$\{(1,a), (1,c)\}$ is not functional because the same first coordinate 1 has two different second coordinates.

6.2 Domain, range and codomain

Definitions

For R from X to Y : $\text{dom}(R) = \{x \mid \text{for some } y \in Y, (x,y) \in R\}$; $\text{ran}(R) = \{y \mid \text{for some } x \in X, (x,y) \in R\}$; the codomain is the declared target set Y .

Example

If $R = \{(2,3), (3,4), (5,100)\}$ is a relation from Z to Z , $\text{dom}(R) = \{2,3,5\}$, $\text{ran}(R) = \{3,4,100\}$, while the codomain is all of Z .

Range $\neq$ codomain in general

A function does not need to hit every element of its codomain. “Range equals codomain” is the extra property called surjectivity, which is handled in Unit 7.

6.3 Function

Definition - function from A to B

A binary relation $f \subseteq A \times B$ is a function $f: A \rightarrow B$ iff (1) f is functional, and (2) $\text{dom}(f) = A$. Equivalently: every $x \in A$ appears exactly once as a first coordinate.

The two-test method

To prove a relation from A to B is a function: TEST 1 functional? TEST 2 domain exactly A? Both must be yes.

Functional but not a function

$S = \{(1, c)\}$ from $A = \{1, 3\}$ to $B = \{a, c\}$ is functional, but $\text{dom}(S) = \{1\} \neq A$. Therefore S is not a function $A \rightarrow B$.

6.4 Formal function proof: $y=x+5$

Test 1 - functional

Assume $(x, y) \in R$ and $(x, z) \in R$. Then $y = x + 5$ and $z = x + 5$, so $y = z$. Hence R is functional.

Test 2 - domain

$\text{dom}(R) = \{x \mid \text{for some } y \in \mathbb{Z}, y = x + 5\}$. For every integer x, $x + 5$ is an integer, so $\text{dom}(R) = \mathbb{Z}$.

Conclusion

Both tests hold, so $R: \mathbb{Z} \rightarrow \mathbb{Z}$ is a function.

6.5 Function notation and images

Notation

$f: A \rightarrow B$ means “f is a function from A to B”. If $(a, b) \in f$, write $f(a) = b$. We say b is the image of a under f, or a maps to b.

People and cities

If $g = \{(Mary, Memel), (Thabo, Ibhayi), (Shawren, Durban)\}$, then $g(Mary) = \text{Memel}$, $g(Thabo) = \text{Ibhay}$ and $g(Shawren) = \text{Durban}$.

Input-output picture

A function is a rule that gives one unambiguous destination $f(a)$ for every allowed input a in the domain.

6.6 Computing examples

Compiler as function - conceptual model

The guide asks you to think of a compiler as transforming a source program (input) into its corresponding object program (output). This illustrates the “one input $\rightarrow$ one determined output” view of functions.

Truncation

The guide treats trunc as a function $\mathbb{R} \rightarrow \mathbb{Z}$ that deletes the fractional part: $\text{trunc}(3.78) = 3$, $\text{trunc}(5) = 5$, $\text{trunc}(-7.22) = -7$.

Polynomial relation

Define $(x,y) \in f$ iff $y=5x^2+3$ on $\mathbb{R}$. Functional: the same x produces the same value $5x^2+3$. Domain: for every real x , $5x^2+3$ is real. Therefore $f:\mathbb{R} \rightarrow \mathbb{R}$ is a function.

6.7 A function's inverse relation need not be a function**Guide example**

$S=\{(4,a),(5,b),(6,a)\}$ from $X=\{4,5,6\}$ to $Y=\{a,b,c\}$ is a function: every input appears once. But $S^{-1}=\{(a,4),(b,5),(a,6)\}$ is not functional because a is paired with both 4 and 6.

Why Unit 7 matters

A function has an inverse function only under stronger conditions. Unit 7 introduces injective/bijective functions, which prevent this collision problem.

6.8 Activity 6-13 - the exam checklist**Given a relation R from A to B , do exactly this**

(i) Prove or disprove that R is functional. (ii) Determine $\text{dom}(R)$ from the domain definition. (iii) Compare $\text{dom}(R)$ with A . Only if R is functional and $\text{dom}(R)=A$ may you conclude $R:A \rightarrow B$ is a function.

6.9 Activity 6-14 - full worked solutions**1. Give five functions $A=\{1,2,3,4\} \rightarrow B=\{a,b,c\}$**

$$f_1=\{(1,a),(2,a),(3,a),(4,a)\}$$

$$f_2=\{(1,a),(2,b),(3,c),(4,a)\}$$

$$f_3=\{(1,b),(2,b),(3,b),(4,c)\}$$

$$f_4=\{(1,c),(2,a),(3,c),(4,b)\}$$

$$f_5=\{(1,a),(2,c),(3,b),(4,c)\}$$

Added counting shortcut

If $|A|=m$ and $|B|=n$, then there are n^m functions $A \rightarrow B$: each of m inputs independently chooses one of n outputs. Here there are $3^4=81$ possible functions. This shortcut is derived directly from the function definition.

2. All functions $\{a,b\} \rightarrow \{5,6,7\}$

$$\{(a,5),(b,5)\} \quad \{(a,5),(b,6)\} \quad \{(a,5),(b,7)\}$$

$$\{(a,6),(b,5)\} \quad \{(a,6),(b,6)\} \quad \{(a,6),(b,7)\}$$

$$\{(a,7),(b,5)\} \quad \{(a,7),(b,6)\} \quad \{(a,7),(b,7)\}$$

Count check

$3^2=9$ functions, matching the nine listed above.

3. Three functions from $A \times A$ to B , $A=\{a,b\}$, $B=\{5,6,7\}$

$A \times A=\{(a,a),(a,b),(b,a),(b,b)\}$. Any valid function must assign exactly one value in B to each of these four inputs.

$$f_1 = \{((a,a),5),((a,b),5),((b,a),5),((b,b),5)\}$$

$$f_2 = \{((a,a),5),((a,b),6),((b,a),7),((b,b),5)\}$$

$$f_3 = \{((a,a),7),((a,b),6),((b,a),6),((b,b),7)\}$$

4. Relation on $A = \{1, 2, 3, \{1\}, \{2\}\}$

$$R = \{(1, \{1\}), (1, 3), (2, \{1\}), (2, \{2\}), (\{1\}, 3), (\{2\}, \{1\})\}.$$

(a) Is R a function $A \rightarrow A$?

No. It is not functional because 1 has outputs $\{1\}$ and 3, and 2 has outputs $\{1\}$ and $\{2\}$. Also $\text{dom}(R) = \{1, 2, \{1\}, \{2\}\}$, which misses 3. Either failure is enough.

(b) Is $\text{range} = \text{codomain}$?

$\text{ran}(R) = \{3, \{1\}, \{2\}\}$. The codomain is $A = \{1, 2, 3, \{1\}, \{2\}\}$. They are not equal.

5. Complement, union and intersection as functions on $P(A)$

(a) $f(x) = x'$

For every $x \in P(A)$, the complement x' is one uniquely determined subset of A , so f is functional. Every $x \in P(A)$ has a complement, so $\text{dom}(f) = P(A)$. Hence $f: P(A) \rightarrow P(A)$ is a function.

(b) $g(u, v) = u \cup v$

For each ordered pair $(u, v) \in P(A) \times P(A)$, the union $u \cup v$ is unique and lies in $P(A)$. Therefore g is functional and its domain is $P(A) \times P(A)$.

(c) $h(u, v) = u \cap v$

Exactly the same structure: $u \cap v$ is unique and is a subset of A . Thus h is functional with domain $P(A) \times P(A)$.

6. Determine which relations are functions

Relation	Function from X to Y ?	Reason
(a) $y = x$ on $\mathbb{Z}$	Yes	Each integer x has exactly one output x ; domain is $\mathbb{Z}$.
(b) $y = x + 1$ on $\mathbb{Z}$	Yes	Unique integer output for every integer x .
(c) $y = 3 - x$ on $\mathbb{Z}$	Yes	Unique integer output for every integer x .
(d) $y = \sqrt{x}$, $X = Y = \mathbb{Z}$	No	Many integers have no integer positive square root; domain is not all $\mathbb{Z}$.
(e) $y^2 = x$ on $\mathbb{Z}$	No	For $x = 1$, $y = 1$ and $y = -1$: not functional; negatives also have no integer output.
(f) $x^2 + y^2 = 1$ on $\mathbb{R}$	No	At $x = 0$, $y = 1$ and $y = -1$, so not functional; $ x > 1$ also has no output.

7. $y=x-1$ on $\mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ **Answer**

No. For $x=1$, the rule gives $y=0$, but $0 \notin \mathbb{Z}^+$ in this guide. Therefore 1 has no output in the codomain, so the domain is not all $\mathbb{Z}^+$. The relation is functional on the inputs that do appear, but it is not a function $\mathbb{Z}^+ \rightarrow \mathbb{Z}^+$.

8. How many equivalence relations on $\{a,b,c\}$ are also functions $A \rightarrow A$?**Answer: 1**

Every equivalence relation is reflexive, so it contains $(a,a), (b,b), (c,c)$. A function may not contain any second pair with the same first coordinate. Therefore a functional equivalence relation can contain no additional pairs. Only the identity relation is both an equivalence relation and a function $A \rightarrow A$.

9(a). How many weak partial orders on A are functions $A \rightarrow A$?**Answer: 1**

Reflexivity again forces all three diagonal pairs. Functionality then forbids every additional pair. The identity relation is reflexive, antisymmetric and transitive, so exactly one weak partial order is a function.

9(b). How many strict partial orders on A are functions $A \rightarrow A$?**Answer: 0**

A function $A \rightarrow A$ on a finite nonempty set gives every element an outgoing edge. Following outputs must eventually create a cycle. A strict partial order is irreflexive and transitive, so cycles are impossible: a cycle would force a self-relation by transitivity. Therefore no strict partial order on $\{a,b,c\}$ can be a function $A \rightarrow A$.

7. Integrated classification method

7.1 Relation-type decision table

Question	If YES	If NO
Is every (x,x) present?	Possible weak order or equivalence relation.	Check irreflexive if looking for a strict order.
Is every (x,x) absent?	Possible strict order.	Not a strict order.
Is it antisymmetric?	Still possible order relation.	Not any of the four COS1501 order types.
Is it transitive?	Still possible order/equivalence.	Not an order relation or equivalence relation.
Is it symmetric?	With reflexive + transitive $\rightarrow$ equivalence.	May still be an order relation.
Are all distinct elements comparable?	Partial order upgrades to total/linear.	It remains partial if the other order properties hold.
Does each first coordinate have at most one output?	Functional relation.	Not functional, therefore not a function.
Is domain exactly the declared source A?	Functional + this $\rightarrow$ function $A \rightarrow B$.	Not a function $A \rightarrow B$.

7.2 One relation can belong to more than one category

Identity relation on A

$I_A = \{(x,x) \mid x \in A\}$ is reflexive, symmetric, antisymmetric and transitive. Therefore it is an equivalence relation and a weak partial order. If A has more than one element, it is not total because distinct elements are incomparable. It is also a function $A \rightarrow A$.

Do not force one label

When a question asks “what kind of relation is this?”, state every relevant classification supported by the properties, unless the question explicitly asks for the most specific order type.

7.3 Fast property fingerprints

Fingerprint	Likely classification
Reflexive + antisymmetric + transitive	Weak partial order
Irreflexive + antisymmetric + transitive	Strict partial order
Weak/strict partial + trichotomy	Weak/strict total order
Reflexive + symmetric + transitive	Equivalence relation
At most one output for each occurring first coordinate	Functional relation
Functional + domain = declared input set	Function

8. Exam traps, memory sheet and practice

8.1 Common exam traps

Trap 1 - confusing symmetric and antisymmetric

Symmetric demands reverse pairs. Antisymmetric forbids reverse pairs only for distinct elements.

Trap 2 - calling a functional relation a function

A missing input is allowed in a functional relation, but not in a function $A \rightarrow B$. Check $\text{dom}(f) = A$.

Trap 3 - checking range instead of domain

For function status, the key set equality is $\text{domain} = \text{source}$. Range may be smaller than the codomain.

Trap 4 - proving a universal property by examples

Examples are intuition only. Use arbitrary variables for a general proof.

Trap 5 - trying to disprove with many cases

One correct counterexample is enough.

Trap 6 - forgetting transitive shortcuts

Whenever xRy and yRz , search for xRz . Missing one shortcut kills transitivity.

Trap 7 - thinking partial means “missing loops”

Partial is about possible incomparability. Weak/strict tells you whether loops are required or forbidden.

Trap 8 - equivalence classes that overlap

If two equivalence classes share even one element, they must be the same class.

Trap 9 - partition with an empty part

A partition's parts must all be nonempty.

Trap 10 - partition that overlaps or misses elements

Parts must be pairwise disjoint and their union must be the whole set.

Trap 11 - mixing tuple coordinates

n -tuples are ordered. Changing coordinate positions changes the record meaning.

Trap 12 - assuming f^{-1} is automatically a function

The inverse relation can give one input multiple outputs. Unit 7 studies when inverse functions exist.

8.2 Final memory sheet

Concept	Must know instantly
Weak partial	Reflexive + antisymmetric + transitive
Strict partial	Irreflexive + antisymmetric + transitive
Weak total	Weak partial + trichotomy
Strict total	Strict partial + trichotomy
Equivalence	Reflexive + symmetric + transitive
Equivalence class	$[x] = \{y \in A \mid xRy\}$
Partition	Nonempty parts, disjoint, union=A
n-ary relation	Subset of $A_1 \times \dots \times A_n$
Functional	Same first coordinate cannot have two different outputs
Function $A \rightarrow B$	Functional and $\text{dom}(f) = A$
Range	Actual second-coordinate values
Codomain	Declared target set B

8.3 Practice test

- Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$. Is R a weak partial order? Is it total?
- Let $S = \{(1, 2), (2, 3), (1, 3)\}$ on $\{1, 2, 3\}$. Classify S as precisely as possible.
- On $\mathbb{Z}$, xRy iff $x - y$ is divisible by 5. Which special kind of relation is R ? Give the equivalence classes.
- Explain why examples cannot prove transitivity on $\mathbb{Z}$.
- Give one counterexample showing divisibility on $\mathbb{Z}^+$ is not a total order.
- Let $A = \{a, b, c\}$. Is $\{\{a, b\}, \{c\}\}$ a partition? Construct the corresponding equivalence relation.
- Is $\{\{a, b\}, \{b, c\}\}$ a partition? State the failed condition.
- What is a 4-ary relation? Give one computing-style 4-tuple.
- From $A = \{1, 2, 3\}$ to $B = \{a, b\}$, is $\{(1, a), (2, b)\}$ functional? Is it a function $A \rightarrow B$?
- Determine whether $\{(1, a), (1, b), (2, a), (3, b)\}$ is a function $\{1, 2, 3\} \rightarrow \{a, b\}$.
- Prove $f = \{(x, y) \in \mathbb{R}^2 \mid y = 2x - 7\}$ is a function $\mathbb{R} \rightarrow \mathbb{R}$.
- If an equivalence class $[x]$ intersects $[y]$, what can you conclude?
- Explain the difference between a weak order and a strict order.
- Explain the difference between a partial and total order.
- How many functions are there from a 3-element set to a 2-element set?

8.4 Practice test answers

- Yes: reflexive, antisymmetric and transitive. Not total because 1 and 3 (also 2 and 3) are incomparable.
- Strict total order: irreflexive, antisymmetric, transitive and all distinct elements comparable.
- Equivalence relation. Five classes $[0], [1], [2], [3], [4]$ modulo 5.
- Transitivity is a universal claim about every possible x, y, z . Finitely many examples cannot cover infinitely many triples.

5. 2 and 3: neither divides the other, so they are incomparable.
6. Yes. $R=\{(a,a),(a,b),(b,a),(b,b),(c,c)\}$.
7. No. The parts overlap in b, so pairwise disjointness fails.
8. A set of ordered 4-tuples, a subset of $A_1 \times A_2 \times A_3 \times A_4$. Example: (student,module,mark,year).
9. Functional yes; function no, because input 3 is missing and $\text{dom} \neq A$.
10. No. Input 1 has two outputs, so the relation is not functional.
11. Functional: if $y=2x-7$ and $z=2x-7$, then $y=z$. Domain: every real x gives real $2x-7$, so $\text{dom}=\mathbb{R}$. Thus $f:\mathbb{R} \rightarrow \mathbb{R}$.
12. They are the same class: $[x]=[y]$. Distinct equivalence classes are disjoint.
13. Weak is reflexive; strict is irreflexive (with the other required order properties retained).
14. Partial may contain incomparable distinct elements; total/linear requires every pair of distinct elements to be comparable.
15. $2^3=8$ functions.

8.5 Completion checklist

- ☐ Define all four order relations from memory.
- ☐ Test reflexive / irreflexive / symmetric / antisymmetric / transitive / trichotomy correctly.
- ☐ Explain partial versus total and weak versus strict.
- ☐ Use general proofs instead of examples for universal claims.
- ☐ Use a counterexample correctly.
- ☐ Explain and use brute force on a small finite set.
- ☐ Explain reductio ad absurdum.
- ☐ Define an equivalence relation and find its equivalence classes.
- ☐ Use representatives and modulo classes.
- ☐ State and apply all four parts of Theorem 6.1.
- ☐ Test all three partition conditions.
- ☐ Construct an equivalence relation from a partition.
- ☐ Define n-ary relations and connect them to database tuples.
- ☐ Distinguish functional relation from function.
- ☐ Determine domain, range and codomain.
- ☐ Prove a relation is a function using "functional + domain=source".
- ☐ Use $f:A \rightarrow B$ and $f(a)=b$ notation.
- ☐ Explain why an inverse relation may not be a function.
- ☐ Solve Activities 6-4, 6-5, 6-7, 6-8, 6-10, 6-12 and 6-14 without looking at the notes.

Final standard

If you can perform every checklist item from an unfamiliar relation, not just repeat the definitions, you have covered the full conceptual and procedural scope of COS1501 Study Unit 6.

9. Unit 6 vocabulary glossary

Term	Meaning in Unit 6
Weak partial order	Reflexive, antisymmetric and transitive order relation.
Strict partial order	Irreflexive, antisymmetric and transitive order relation.
Comparable	Two distinct elements x, y are comparable if xRy or yRx .
Trichotomy	The comparability condition used to make a partial order total/linear.
Weak total / linear order	Weak partial order plus trichotomy.
Strict total / linear order	Strict partial order plus trichotomy.
Counterexample	One case that shows a universal statement is false.
Brute force	Proof by checking all possibilities in a small finite situation.
Reductio ad absurdum	Proof by assuming the opposite and deriving a contradiction.
Equivalence relation	Reflexive, symmetric and transitive relation.
Equivalence class $[x]$	Set of all elements equivalent to x .
Representative	Chosen element used to name an equivalence class.
Partition	Collection of nonempty, pairwise disjoint subsets whose union is the whole set.
Ordered n -tuple	An ordered sequence $(x_1, \dots, x_n)$ of n coordinates.
n -ary relation	A set of n -tuples; subset of $A_1 \times \dots \times A_n$.
Functional relation	Each occurring first coordinate has exactly one second coordinate.
Domain	First-coordinate values that actually occur.
Range	Second-coordinate values that actually occur.
Codomain	Declared target set of the relation/function.
Function	Functional binary relation whose domain equals the declared source set.
Image	For $f(a)=b$, b is the image of a .
Maps to	Language describing $a \rightarrow f(a)$.
Inverse relation	Relation formed by reversing every ordered pair.
$P(A)$	Power set of A ; all subsets of A .
modulo	Remainder operation; equivalence classes often group integers by the same remainder.

Source basis

Primary source: supplied UNISA COS1501 study guide, Study Unit 6 “Special kinds of relation”, pp. 83-102. The source’s terminology, section order and official examples/activities were used as the basis. Added explanations, extra examples, solved activities, memory rules and exam notes are original teaching material written for this PDF.